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Evaluating the Impact of Intestinal Bile Salts on Drug Absorption Using PBPK Modeling: Case Studies With Efavirenz, Cinnarizine, and Posaconazole

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ABSTRACT

A high number of poorly soluble compounds are being developed; thus, understanding the factors that influence their absorption is critical. Intestinal bile salts, which facilitate micelle-mediated solubilization, are particularly important for drugs with low solubility and are reported to be highly variable. The aim of this study was to evaluate the effect of luminal bile salt concentrations on the absorption of poorly soluble compounds, using efavirenz, cinnarizine, and posaconazole as examples. Physiologically-based pharmacokinetic (PBPK) models were developed and validated using the Simcyp Simulator. Sensitivity analyses were performed to assess the impact of bile salts and other gastrointestinal parameters on drug absorption. Simulations revealed that drug absorption in the fasted state was most sensitive to bile salt concentrations compared to other gastrointestinal parameters such as luminal pH, fluid volumes, and gastric emptying. The findings indicate that efavirenz, cinnarizine, and posaconazole exhibit high micelle-mediated solubility, with bile salts playing a critical role in their absorption, particularly in the fasted state. These results highlight the importance of considering bile salt concentrations in PBPK modeling of poorly soluble compounds.

1 | Introduction

Oral drug administration accounts for 90% of the global market share of clinically administered drugs [1]. Whilst this route of administration is the most convenient, limited and variable bioavailability can complicate reaching the desired clinical effect. To reach the systemic circulation, the drug must first be dissolved and then permeate the intestinal wall. Bile salts play a major role in solubilizing drugs via the formation of micelles, which poorly soluble compounds are particularly dependent on. Efforts to enhance metabolic stability have resulted in an increasing number of poorly soluble drugs on the market, further limiting bioavailability, with 90% of new chemical entities being insufficiently water-soluble [2]. According to the Biopharmaceutics Classification System (BCS), class II compounds are characterized by low solubility and high permeability. Given the

increasing number of poorly soluble compounds on the market, physiologically-based pharmacokinetic (PBPK) modeling has become an essential tool for predicting oral drug absorption and aiding in formulation design. PBPK models provide insights into how various physiological factors, such as bile salt concentrations, luminal pH, and intestinal fluid volumes, affect the absorption of poorly soluble drugs.

While many poorly soluble compounds with a positive food effect are administered with food to improve absorption, this is not always feasible. For example, efavirenz is typically administered in the fasted state due to its erratic absorption when given with food [3]. Although bile salt concentrations peak in the fed state, they are continuously secreted throughout the interdigestive migrating motor complex (IMMC) during the fasted state, and their presence extends across the gastrointestinal tract (GIT) [4].

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Study Highlights

- What is the current knowledge on the topic?
 - Bile salts increase solubility for certain lipophilic, poorly soluble drugs. This can be mechanistically captured using PBPK modeling, alongside the complex solubility–permeability interplay.
- What question did the study address?
 - This study aimed to address how sensitive the solubility, dissolution, and absorption of drugs with diverse physicochemical properties (e.g., efavirenz, cinnarizine, and posaconazole) are to intestinal bile salts. The questions the study addresses are: (1) To what extent do bile salts modulate drug solubility across physiological conditions in fasted and fed states? (2) How is solubility–permeability interplay captured?
- What does this study add to our knowledge?
 - By examining efavirenz, cinnarizine, and posaconazole as case studies, this work provides new insights into how variability in bile salt concentrations can explain clinically observed differences in drug absorption. The study also demonstrates the importance of accurately integrating bile salt-dependent solubility and permeability parameters into PBPK models.
- How might this study change drug discovery, development, and/or therapeutics?
 - Many drugs in development are poorly soluble and may have improved absorption due to bile salts. This study demonstrates the utility of PBPK modeling to investigate this, especially in specific populations where bile salt concentrations may differ due to physiological or pathological conditions.

Nevertheless, drug absorption can be problematic in the fasted state, especially when bile salt concentrations fall below the critical micelle concentration (CMC), the threshold required for micelle formation. Furthermore, bile salt concentrations are reported to be reduced in certain specific populations, such as those with hepatic impairment, whereby duodenal bile salts are reduced by approximately two-fold [5].

A recent meta-analysis of bile salt concentrations in healthy adults in both fasted and fed states has been implemented into the Simcyp simulator, accounting for inter-individual variability [6]. This variability means that in some cases, bile salt concentrations may fall below the CMC in certain segments of the GIT, potentially impairing absorption in individuals with lower bile salt levels.

The interplay between solubility and permeability is another critical factor in drug absorption, particularly for poorly soluble compounds [7]. In PBPK modeling, permeability modules that account for regional differences may be considered over directly inputting an in vitro permeability value. The aims of this study were to assess the impact of bile salt concentrations on oral drug absorption for poorly soluble compounds, with

a particular focus on micelle-mediated absorption. Efavirenz, cinnarizine, and posaconazole were selected as BCS II model compounds to investigate how changes in bile salt concentrations, as well as other GI parameters, influence drug absorption in fasted and fed state, accounting for different permeability models.

2 | Methods

2.1 | Compound Selection

The compounds explored in this study were selected based on having high micelle-mediated solubility determined in vitro ($\geq 70\%$) as well as a clinically positive food effect. The micelle-mediated solubility was calculated as the percentage of FaSSIF solubility/total solubility, whereby the FaSSIF solubility is the fasted state simulated intestinal fluid solubility and the total solubility is the sum of the FaSSIF and blank FaSSIF solubility. The blank FaSSIF media is the same composition as FaSSIF but without bile salts. These data are summarized in Table 1.

2.2 | PBPK Model Development

All simulations were performed in Simcyp Version 23 (Certara Predictive Technologies, Sheffield, UK), using the mechanistic Advanced Dissolution Absorption and Metabolism (ADAM) model. PBPK models were developed for efavirenz and cinnarizine. For posaconazole, an existing PBPK model for oral suspension was used and adapted [11]. For all compounds, fasted and fed state models were developed. If the model was deemed fit-for-purpose, it was included in the sensitivity analysis. Since the focus was absorption, this was evaluated based on the simulated maximal concentration ($C_{\max}$) being within 0.8–1.25-fold of the observed value. Additionally, the impact of using the mechanistic effective permeability ($\text{MechP}_{\text{eff}}$) model for permeability input was evaluated. The modeling approach is outlined in Figure 1.

TABLE 1 | Micelle-mediated solubility and clinical food effect for efavirenz, cinnarizine and posaconazole.

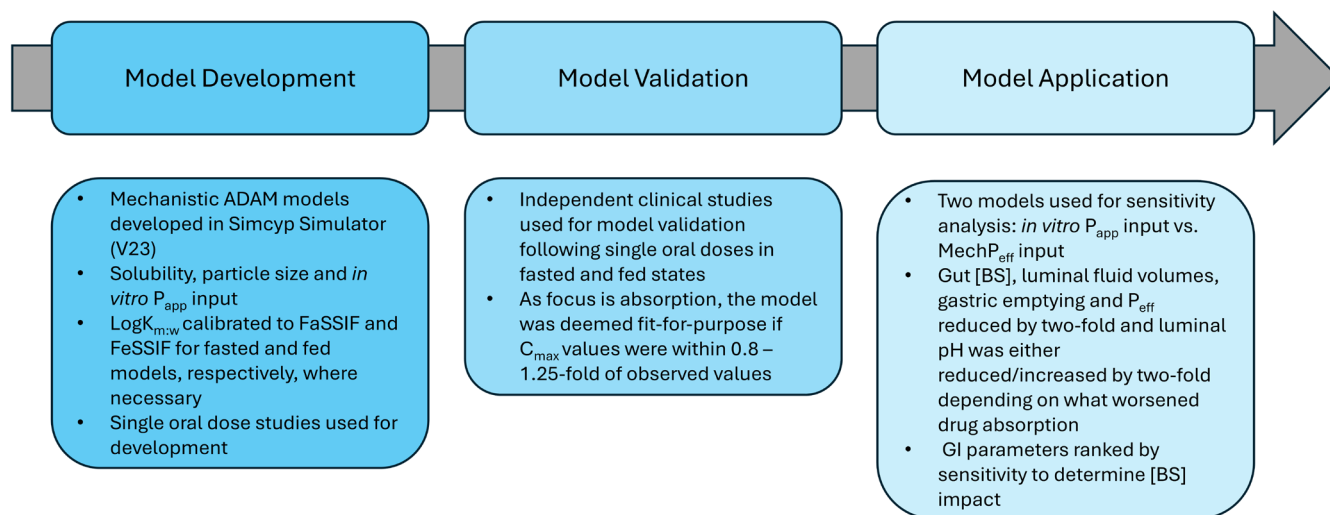
Compound	Micelle-mediated solubility	Clinical food effect
Efavirenz	95.1%	$C_{\max}$ and AUC increased by 22% and 39%, respectively, following high-fat meal [8] and 17% and 51%, respectively, following low-fat meal [8]
Cinnarizine	97.8%	$C_{\max}$ and AUC increased by 28% and 57% following low-fat meal [9]
Posaconazole	78.8%	$C_{\max}$ and AUC increased by 288% and 291%, respectively, following high-fat meal [10]

2.3 | Absorption Input Parameters

The ADAM model handles drug dissolution and transit through different regions of the GIT [12]. Parameterization for the ADAM model is summarized in Figure 2. The solubility inputs for the compounds used in this study are summarized in Table 2. Intrinsic solubility (S_0) was either integrated as an experimental value or back-calculated based on pH solubility, depending on data availability and model performance. Bile

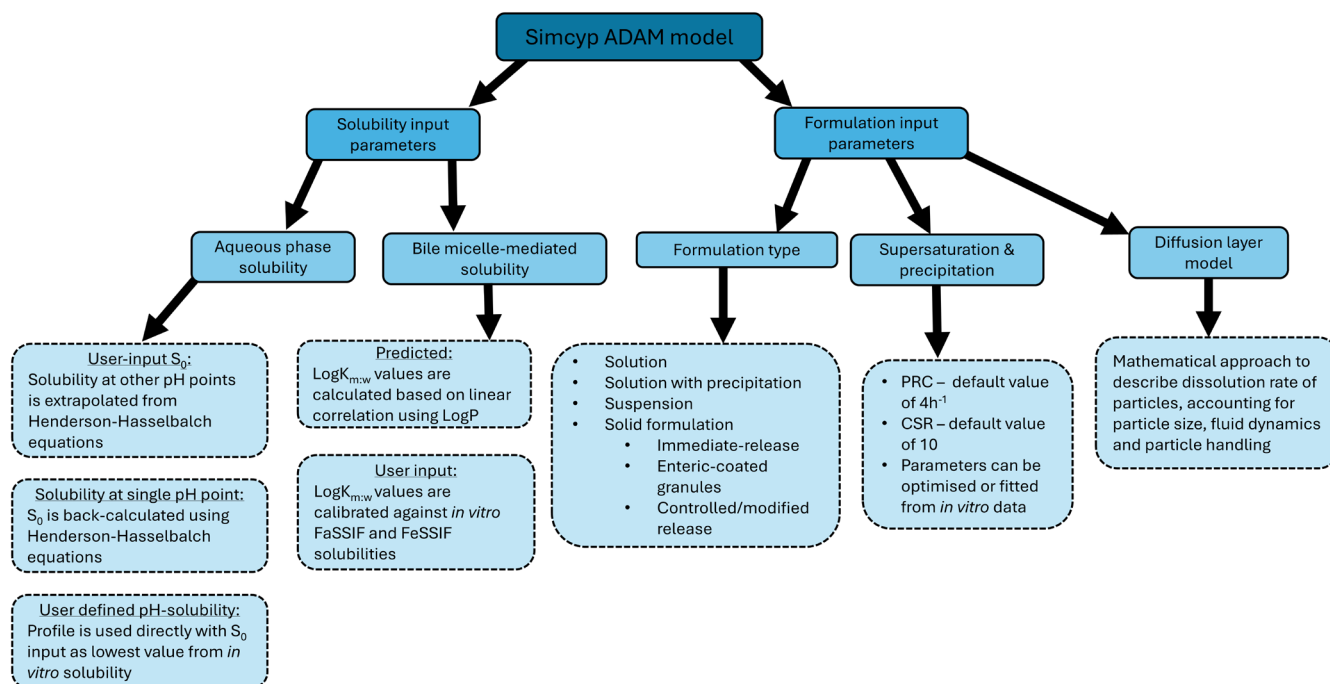
micelle partitioning ($\text{LogK}_{m:w}$) was either predicted or calibrated against reported FaSSIF and fed state simulated intestinal fluid solubilities, depending on solubility predictions. Details on all available solubility data and model parameterization are outlined in the [Supporting Information](#).

Within the ADAM model, total solubility is predicted as the sum of aqueous solubility (from the ionized species) and micelle-mediated solubility (from the unionized species). These are



P_{app} = apparent permeability; $\text{LogK}_{m:w}$ = bile micelle partitioning; FaSSIF = fasted state simulated intestinal fluid; FeSSIF = fed state simulated intestinal fluid; C_{max} = maximal plasma concentration; MechP_{eff} = mechanistic effective permeability; [BS] = bile salt concentrations; GI = gastrointestinal.

FIGURE 1 | Schematic diagram of PBPK modeling strategy.



S_0 = intrinsic solubility; $\text{LogK}_{m:w}$ = bile micelle partitioning; FaSSIF = fasted state simulated intestinal fluid; FeSSIF = fed state simulated intestinal fluid; PRC = precipitation rate constant; CSR = critical supersaturation ratio.

FIGURE 2 | Simcyp ADAM model parameterization for solubility and formulation inputs.

TABLE 2 | Summary of aqueous solubility and bile micelle-mediated solubility input for efavirenz, cinnarizine, and posaconazole PBPK models.

Compound	Solubility input	LogK _{m:w}
Efavirenz	User-input S_0 (experimental)	Calibrated
Cinnarizine	Back-calculated S_0 from single pH-solubility point	Predicted
Posaconazole	User-input S_0 (experimental)	Calibrated

governed by Henderson-Hasselbalch equations, with S_0 and LogK_{m:w} serving as key inputs to estimate the solubility of each species under physiological pH and bile salt concentrations [12].

For the permeability, an in vitro apparent permeability (P_{app}) value can be directly inputted from cell lines, such as adenocarcinoma (Caco-2), or as a predicted value based on quantitative structure–activity relationships (QSAR). Alternatively, the (MechP_{eff}) model offers a more detailed prediction of P_{eff} by considering regional P_{eff} values across the gut [13], allowing for a better representation of the solubility-permeability interplay, especially when drug solubility is mediated by micelle formation. Thus, whilst an in vitro permeability P_{app} value is scaled to a P_{eff} value applied across all GI segments, the MechP_{eff} model considers regional differences.

2.3.1 | Efavirenz

2.3.1.1 | Absorption Model Input Data for Efavirenz. In the Simcyp simulator, the existing compound file for efavirenz was used. Since this uses a first-order absorption model, this was switched to the ADAM model, where solubility and formulation input parameters were implemented. An immediate release (IR) formulation was simulated with an S_0 of 0.00855 mg/mL [14]. As the particle size of the API itself has not been directly reported, the median particle size for a batch bioequivalent to the API [15], was used in the model (particle radius = 1.64 μ m). Solubility in biorelevant media was used to calibrate the LogK_{m:w} for the neutral and ionized species. The final input parameters used for the efavirenz ADAM model are summarized in Table S1.

2.3.2 | Cinnarizine

2.3.2.1 | Model Assumptions and Input Data for Cinnarizine. The solubility of cinnarizine is pH-dependent, with reported solubility values of 0.29 mg/mL at pH 2, 0.017 mg/mL at pH 5 and 0.002 mg/mL at pH 6.5 [16]. An S_0 value has not been reported for cinnarizine. Directly inputting the pH-solubility profile resulted in worse predictions than using a single pH-solubility point (solubility at pH 6.5) to back-calculate S_0 ; thus, this was inputted in the final model. The particle radius of cinnarizine is reported to be 14.5 μ m [17]. A moderate P_{app} value of 4.2×10^{-6} cm/s has been reported for cinnarizine in Caco-2 cell lines [18]. A full PBPK model for cinnarizine was simulated using an ADAM IR tablet formulation. Further details on the model development are provided in the Supporting

Information, and the PBPK model input parameters are shown in Table S2.

2.3.3 | Posaconazole

An existing PBPK model for posaconazole oral suspension was used, based on Bhatnagar et al. [11]. To better capture the prolonged t_{max} , the fraction of unbound drug in the gut ($f_{u_{gut}}$) was modified to equal the f_{u_p} and the free enterocyte was selected as the driving concentration into the portal vein. The LogK_{m:w} values were calibrated to the FaSSiF and FeSSiF solubilities.

2.4 | Sensitivity Analysis

To investigate the impact of bile salts on oral drug absorption, sensitivity analyses were performed around intestinal bile salt concentrations, whereby simulations with 50% reduced bile salt concentrations for each segment were run. This was done to mimic the two-fold reduction in duodenal bile salts reported for hepatic impairment patients. The coefficient of variation (CV) was set to 30% for all segments to keep the concentrations closer to the intended two-fold reductions to reflect the desired physiological changes, while still maintaining some natural variability. This was simulated for both prandial states, and permeability model inputs. To assess sensitivity of bile salts, the same two-fold reduction was applied to other gastrointestinal parameters, including gastric emptying, luminal fluid volumes and luminal pH. For the pH, both increased and decreased scenarios were simulated and the scenario used for the sensitivity ranking was that which resulted in the poorer absorption. The CVs for these parameters were also set to 30% to allow for a more consistent and controlled assessment of parameter reductions. The impact of reduced P_{eff} was also investigated by reducing P_{eff} by 50% for both permeability scenarios, i.e., reducing the overall P_{eff} from P_{app} for the in vitro P_{app} input and reducing P_{eff} for each segment for the MechP_{eff} model input. The fa was used to evaluate the level of sensitivity of each parameter. When switching to the MechP_{eff} model, no other changes were made to the model regardless of how well the absolute PK were recovered. A sensitivity analysis where both bile salt concentrations and P_{eff} were reduced simultaneously was also performed.

In the Simcyp simulator, a CMC of 1 mM is assumed, and below this threshold, solubility enhancement through bile micelle partitioning is not considered. It should be noted that whilst this is the default CMC value, the CMC is variable and dependent on bile salt concentrations and associated counterions [19].

3 | Results

3.1 | Model Development & Validation

3.1.1 | Efavirenz

The area-under-the-curve (AUC) and C_{max} for the fasted efavirenz model were within 0.8–1.25-fold of observed values (Table S4). The PK profile for the Ogburn et al. study [20] was well recovered (Figure S1). The duodenal micelle-mediated

solubility was predicted to be 96.4%. Whilst the fed exposure was overpredicted (AUC_{0-t} fed/fasted ratio of 1.36), the C_{max} for the tablet was well recovered (fed/fasted ratio of 1.01). Whilst tablet and capsule formulations are reported to be bio-equivalent, there are discrepancies in the fed state, whereby a higher C_{max} and lower AUC are reported for the tablet. Whilst the fasted model showed good recovery for both formulations, the fed-state discrepancies in the clinical data could not be mechanistically captured in the model (further details in Section 5 of the [Supporting Information](#)). Thus, the fed model should only be used for exploratory analyses.

For the fasted model validation, most studies administered a 600mg single dose tablet, although Study DMP 266–108 also administered two 300mg tablets. The predicted fa across the fasted studies ranged from 0.54 to 0.67, which aligns with the reported range of 0.40–0.55 [21]. The AUC and C_{max} were all within 0.8–1.25-fold of observed values (Table S5), hence the fasted model was deemed fit-for-purpose. The variability was predicted to be high (C_{max} CVs ranged from 92.9% to 94.7%) across the studies in comparison to the observed variability (C_{max} CVs ranged from 23.0% to 26.3%).

3.1.2 | Cinnarizine

For the cinnarizine fasted model, the concentration-time profile was well recovered (Figure S2) and the AUC and C_{max} were within 0.8–1.25-fold of the observed data for the development study (Table S6). The duodenal micelle-mediated solubility was predicted to be 80.7%. The fed model was well recovered (Figure S3) with AUC and C_{max} within 1.5-fold of the observed data and fed/fasted ratios within 0.8–1.25-fold (Table S6).

For the model validation, two independent clinical studies were simulated. For the Nowacka-Krukowska et al. study [22], the AUC_{inf} was underpredicted (AUC ratio=0.68) but the AUC_{0-t} was within 0.82-fold of the observed value (Table S7). However, both AUC values for the Abouelatta et al. study [23], were within 0.8–12.5-fold of observed values. This is likely due to differences in how AUC was calculated. For the former study, the AUC was calculated using the trapezoidal rule and then extrapolated to infinity, whilst for the latter study, the AUC to infinity is subdivided, where $AUC_{inf} = AUC_{0-24} + C_t/K$, with C_t being the last sample concentration at time t .

3.1.3 | Posaconazole

For the fasted and fed state simulations of posaconazole, a 200mg single dose and a 400mg single dose were administered, respectively. Concentration-time profiles were well recovered for both prandial states (Figures S6 and S7, respectively) and the AUC and C_{max} were within 0.8–1.25-fold of the observed data for the development study. The same was observed for the validation studies (Figure S8 and Table S8). Hence, both fasted and fed state models were deemed fit-for-purpose for the sensitivity analysis. The fasted duodenal micelle-mediated solubility was predicted to be 91.0%.

3.2 | Sensitivity Analysis

The results following a two-fold reduction in intestinal bile salt concentrations, P_{eff} , gastric emptying, luminal fluid volumes and luminal pH (two-fold increase for cinnarizine and posaconazole) are summarized for each compound (Figure 3, Table 3).

3.2.1 | Efavirenz

The baseline fa for efavirenz with in vitro P_{app} is moderate and in alignment with the observed fa (0.40–0.55), whilst the $MechP_{eff}$ model baseline fa is significantly lower (Table 3). The P_{eff} from the in vitro P_{app} input is predicted to be 9.77×10^{-4} cm/s and the $MechP_{eff}$ model predicts a similar overall value of 10.2×10^{-4} cm/s. However, regional P_{eff} predictions were significantly lower. Bile salts are critical for efavirenz absorption in the fasted state, with fa reducing by 41% following two-fold reductions in bile salts (in vitro P_{app}), highlighting the importance of micelle-mediated solubility for this compound. There is a substantial impact on fa when permeability is reduced, especially when using the $MechP_{eff}$ model (–54%), indicating permeability to be a significant barrier to efavirenz absorption, especially when micelle formation is involved. Moderate changes in fa are seen when fluid volumes are reduced (–23% to –27%), suggesting that luminal fluid volumes also play an important role. Changes in gastric emptying are minimal. Since efavirenz does not have pH-dependent solubility, there is no impact with altered luminal pH. Overall, bile salts and permeability are the most influential factors, followed by fluid volumes. In the fed state, bile salts are less significant, with fluid volumes and permeability being crucial for the $MechP_{eff}$ model. Following simultaneous reductions in bile salts and permeability, there is a summative impact for both fasted models and the fed $MechP_{eff}$ model, but not much impact for the fed in vitro P_{app} model (Table 4).

3.2.2 | Cinnarizine

Cinnarizine has a baseline predicted fa of 0.64 for the in vitro P_{app} model and 0.78 and for the $MechP_{eff}$ model. Whilst the P_{eff} from the in vitro P_{app} input is predicted to be 1.22×10^{-4} cm/s, the $MechP_{eff}$ model predicts a significantly higher overall value of 18.9×10^{-4} cm/s.

Bile salts have a moderate impact on cinnarizine absorption in the fasted state using in vitro P_{app} , with fa decreasing by 19% following two-fold reductions in bile salts, but not much impact with the $MechP_{eff}$ model. Reducing P_{eff} causes a moderate decrease in fa (–20% to –36%), indicating that permeability is another key factor. Both reducing/increasing luminal pH resulted in increased fa, although increased pH resulted in a lower fa overall and so this was reported (+28% in vitro P_{app}). In the fed state, bile salts are more significant with the $MechP_{eff}$ model (–15%). Overall, bile salts play a less significant role for cinnarizine, despite high micelle-mediated solubility. Following simultaneous reductions in bile salts and permeability, there is a summative impact for both fasted models but not much impact for the fed models compared with reduced P_{eff} alone (Table 4).

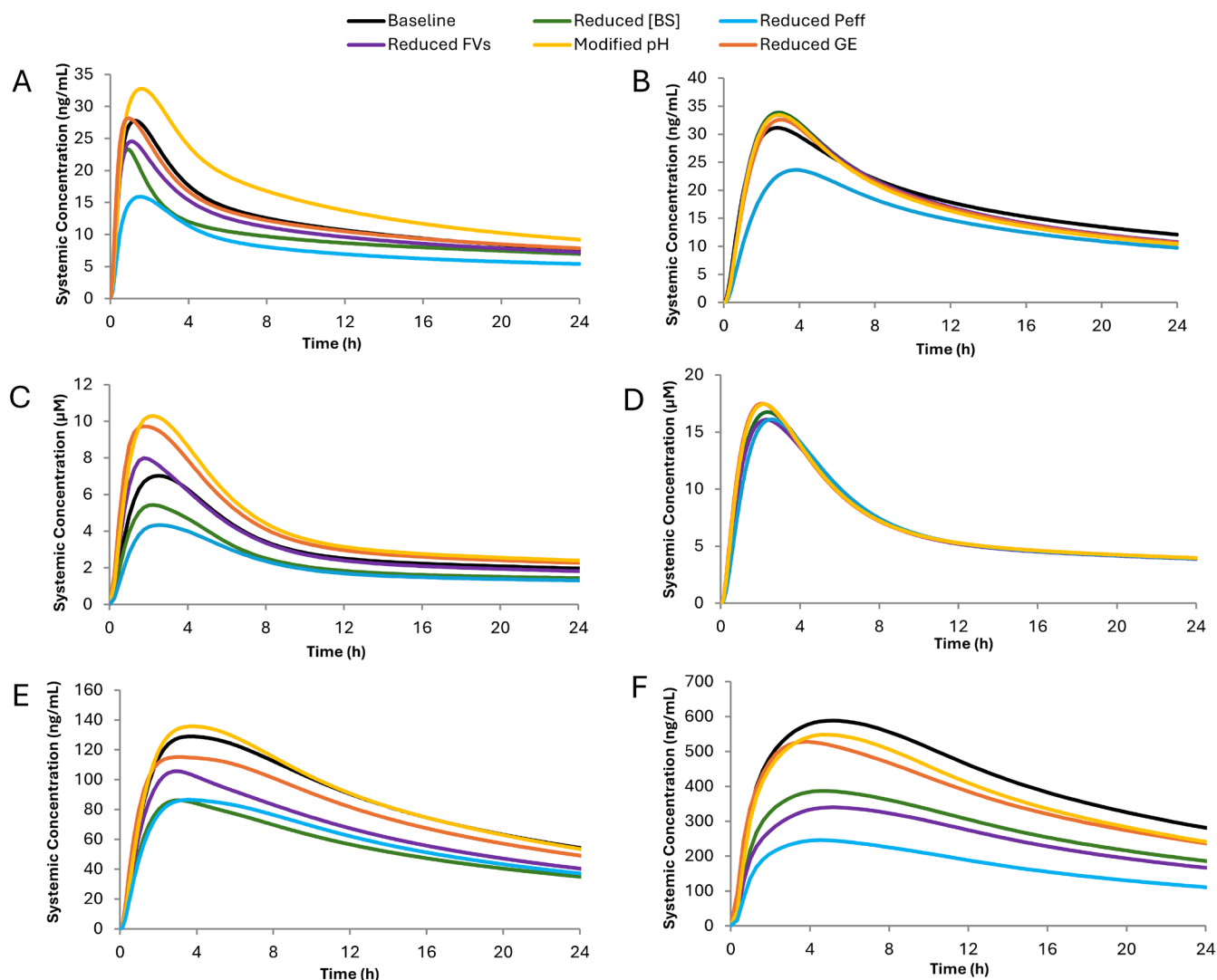


FIGURE 3 | Plasma concentration-time profiles for fasted (A) and fed (B) efavirenz, fasted (C) and fed (D) cinnarizine, and fasted (E) and fed (F) posaconazole following a 50% reduction in GI parameters/ P_{eff} (or increase for efavirenz and cinnarizine pH).

3.2.3 | Posaconazole

Very low f_a values were predicted for posaconazole, even in the fed state. This was reduced further with the Mech P_{eff} model. Whilst the P_{eff} from the in vitro P_{app} input is predicted to be 6.41×10^{-4} cm/s, the Mech P_{eff} model predicts an overall value of 8.48×10^{-4} cm/s. Posaconazole was more sensitive to reduced P_{eff} than reduced bile salt concentrations for both permeability models and both prandial states. Bile salts significantly affected posaconazole absorption, especially in the fed state (–35% reduction in f_a). However, permeability had the most dramatic effect on f_a , particularly with the Mech P_{eff} model (–50% change in both fasted and fed states). This suggests that permeability is a critical limiting factor for posaconazole, especially in micelle-mediated solubility. Whilst bile salts predominate for the fasted in vitro P_{app} model, pH and fluid volumes play a more significant role for the Mech P_{eff} model and fluid volumes for the fed models. Following simultaneous reductions in bile salts and permeability, there is a summative impact for both fasted models but not much impact for the fed models compared with reduced P_{eff} alone (Table 4).

4 | Discussion

In this study, three poorly soluble compounds, efavirenz, cinnarizine, and posaconazole, were selected to investigate their sensitivity to intestinal bile salt concentrations. These compounds appeared to have significant micelle-mediated solubility in vivo, based on two criteria: (1) in vitro solubility studies showing significant micelle-mediated solubility ($\geq 70\%$) and (2) a positive food effect observed in clinical studies. Fit-for-purpose models were developed for efavirenz, cinnarizine, and posaconazole in fasted and fed states. The findings indicate that sensitivity to bile salts is dependent on permeability model input and prandial state and is compound-specific. Whilst bile salts had less impact for cinnarizine and efavirenz in the fed state, there was moderate impact for posaconazole.

4.1 | Compound-Specific Findings

Efavirenz (a weak acid with pK_a 10.2 and a $\log P$ of 4.02) exhibited significant sensitivity to bile salt concentrations, especially

TABLE 3 | Relative change in compound fa following two-fold reduction in gut [BS], gastric emptying, luminal fluid volumes, and P_{eff} and two-fold increased/reduced luminal pH for in vitro P_{app} permeability input versus Mech P_{eff} permeability input for fasted and fed states.

Compound	Prandial state	Permeability model input	% change in fa							Sensitivity ranking for GI parameters
			Baseline fa	Reduced [BS]	Reduced GE	Reduced FVs	Modified pH	Reduced P _{eff}		
Efavirenz	Fasted	In vitro P _{app}	0.56	-41	-5	-27	0	-34	[BS] > FVs > GE > pH	
		MechP _{eff}	0.13	-23	-8	-23	0	-54	[BS] & FVs > GE > pH	
	Fed	In vitro P _{app}	1.00	-1	0	-2	0	-1	FVs > [BS] > GE & pH	
		MechP _{eff}	0.38	-8	0	-45	0	-53	FVs > [BS] > GE & pH	
Cinnarizine	Fasted	In vitro P _{app}	0.64	-19	0	-8	+28	-36	pH > [BS] > FVs > GE	
		MechP _{eff}	0.78	-4	0	-5	+5	-28	FVs & pH > [BS] > GE	
	Fed	In vitro P _{app}	0.99	-3	-4	-4	-7	-20	pH > FVs & GE > [BS]	
		MechP _{eff}	0.86	-15	+15	+1	-3	-27	[BS] & GE > pH > FVs	
Posaconazole	Fasted	In vitro P _{app}	0.12	-33	-17	-25	0	-33	[BS] > FVs > GE > pH	
		MechP _{eff}	0.04	0	0	-25	+25	-50	pH & FVs > [BS] & GE	
	Fed	In vitro P _{app}	0.26	-35	-15	-38	0	-42	FVs > [BS] > GE > pH	
		MechP _{eff}	0.06	-17	-17	-33	0	-50	FVs > [BS] & GE > pH	

Abbreviations: [BS], bile salt concentrations; Fa, fraction of drug absorbed; FVs, fluid volumes; GE, gastric emptying; P_{eff} , effective permeability.

TABLE 4 | Relative change in compound fa following simultaneous two-fold reduction in gut [BS] and P_{eff} for in vitro P_{app} permeability input versus $\text{Mech}P_{\text{eff}}$ permeability input for fasted and fed states.

Compound	Prandial state	Permeability model input	Baseline fa	% change in fa with reduced [BS] & P_{eff}
Efavirenz	Fasted	In vitro P_{app}	0.56	−63
		$\text{Mech}P_{\text{eff}}$	0.13	−77
	Fed	In vitro P_{app}	1.00	−7
		$\text{Mech}P_{\text{eff}}$	0.38	−66
Cinnarizine	Fasted	In vitro P_{app}	0.64	−52
		$\text{Mech}P_{\text{eff}}$	0.78	−40
	Fed	In vitro P_{app}	0.99	−19
		$\text{Mech}P_{\text{eff}}$	0.86	−27
Posaconazole	Fasted	In vitro P_{app}	0.12	−67
		$\text{Mech}P_{\text{eff}}$	0.04	−75
	Fed	In vitro P_{app}	0.26	−7
		$\text{Mech}P_{\text{eff}}$	0.06	−66

Abbreviations: [BS], bile salt concentrations; Fa, fraction of drug absorbed; P_{eff} , effective permeability.

when permeability was modeled using in vitro P_{app} . This result suggests that micelle-mediated solubility is critical for efavirenz absorption in the fasted state. Interestingly, when using the $\text{Mech}P_{\text{eff}}$ model, fa decreased more drastically with reduced permeability. This highlights that permeability is a key limiting factor for efavirenz, particularly when micelle formation is considered. Changes in gastric emptying and luminal pH had minimal impact on absorption, consistent with efavirenz's lack of pH-dependent solubility. However, luminal fluid volumes had a moderate impact, demonstrating that while bile salts and permeability are the primary influencers of efavirenz absorption, fluid volumes still play a role.

Cinnarizine (a base with pKa 1.95 and 7.47 and a LogP of 5.6) showed a slightly different sensitivity profile for the other gastrointestinal parameters, but like efavirenz, bile salts predominated. The sensitivity to pH changes was also significant, especially for the in vitro P_{app} model, capturing cinnarizine's pH-dependent solubility. Cinnarizine's higher baseline fa in both permeability models, particularly in the $\text{Mech}P_{\text{eff}}$ model, suggests relatively high permeability, which mitigates the impact of changes in other parameters like gastric emptying and fluid volumes. The sharp reduction in fa with decreased bile salt concentrations highlights the critical role of micelle formation in cinnarizine absorption.

Posaconazole (a base with pKa 3.6 and 4.6 and a Log P of 4.6) demonstrated permeability to be a more critical factor than bile salts, especially when modeled using $\text{Mech}P_{\text{eff}}$. The 50% reduction in fa with reduced P_{eff} shows that, despite its low aqueous solubility, permeability is the key limiting factor for posaconazole absorption. Even in the fed state, posaconazole's fa remained low, indicating that its absorption is constrained by both solubility and permeability limitations. Interestingly, luminal fluid volumes had a substantial impact in the fed state, possibly due to posaconazole's formulation characteristics.

4.2 | Model Limitations and Data Constraints

While the models developed reasonably capture the solubility–permeability interplay under fasted and fed conditions, the AUC for efavirenz in the fed state was overpredicted. Efavirenz was not sensitive to precipitation or solubility changes. However, given the clinical study has a moderate sample size ($n = 22$) with relatively low variability (CV for $\text{AUC}_{\text{inf}} = 23.7\%$), there are likely formulation-specific differences not captured in the model. The reported adverse events relating to the central nervous system have been linked to the high C_{max} in the fed state [8]. Given the C_{max} is well recovered, the fed model was deemed fit-for-purpose.

4.3 | Permeability Model Considerations

In this study, both in vitro P_{app} and $\text{Mech}P_{\text{eff}}$ approaches were explored to evaluate the impact of permeability modeling on bile salt sensitivity and drug absorption. While the $\text{Mech}P_{\text{eff}}$ model tended to underpredict fa, this is not necessarily indicative of a deficiency in the $\text{Mech}P_{\text{eff}}$ model itself but rather reflects differences in assumptions and physiological representation between the two modeling approaches. The $\text{Mech}P_{\text{eff}}$ model incorporates intestinal surface area, segmental permeability, and effective unstirred water layer thickness, offering a more mechanistic perspective on regional absorption. However, when transitioning between permeability models, it is common practice to adjust other parameters (e.g., gut physiology) to better capture the observed fa. In this study, no such adjustments were made as the goal was to compare sensitivity to bile salt-mediated solubility using two different permeability approaches. While model refinement could improve alignment with observed fa values, it would confound the direct comparison between input methods. As such, the current study should be interpreted in terms of *relative* trends

and physiological insights, rather than *absolute* predictive performance. In most cases, the MechP_{eff} model was found to be less sensitive to bile salts than the in vitro P_{app} model, likely due to the reduced solubility being compensated for by the increased permeability. MechP_{eff} models with further refinement (i.e., other absorption changes) and subsequent sensitivity analyses were performed for comparison. Given this is beyond the scope of this study, results are included in the [Supporting Information](#) (Tables S9–S12).

4.4 | Implications for Hepatic Impairment

Bile salt concentrations are known to vary significantly among individuals, including healthy subjects [6]. They are further reduced in certain special populations, such as hepatic impairment patients, whereby duodenal concentrations are reported to be reduced by approximately two-fold [5, 24, 25]. Thus, these patients are likely to have bile salt concentrations below the CMC, which could impair the absorption of bile salt-dependent drugs. PBPK modeling offers a valuable tool to evaluate how such variability in bile salts could affect drug absorption, particularly for BCS II compounds.

A clinical study in hepatic impairment for entrectinib reported reduced absorption in moderate and severe impairment in the fed state [26]. In this study, PBPK modeling was used to compare predictions to observed data, where the MechP_{eff} model was used. To recover the C_{max}, the luminal bile salt concentrations were reduced by 2-, 6- and 8.7-fold for mild, moderate and severe impairment, going against in vitro findings. It should be noted that entrectinib has a negligible food effect [27], thus would not be expected to be highly sensitive to bile salts. Given the MechP_{eff} model was used, there may be a counterbalance between increased solubility and reduced permeability, allowing a drastic reduction in bile salts to be compromised by the increased gut permeation. Whilst the MechP_{eff} model is more mechanistic, assumptions around micelle-bound drug may not be physiologically representative for poorly soluble compounds.

In this study, a two-fold reduction in luminal bile salts has resulted in a significant reduction in fa for compounds with varying baseline fa values and varying extent of food effect. Given the available in vitro data and the simulations performed here, reductions of up to nearly 9-fold in hepatic impairment are likely to be physiologically unrealistic. Whilst the empirical approach described by Ozbey et al. [26] can help gain an understanding of bile salt sensitivity, there are likely other mechanisms involved in the absorption impairment of entrectinib in hepatic impairment.

5 | Conclusions

This study highlights the critical role of bile salts in the absorption of poorly soluble, micelle-mediated compounds. The sensitivity of fa to changes in permeability and solubility parameters emphasizes the complex interplay of drug absorption. The findings also highlight the potential that PBPK modeling has in capturing absorption variability and informing formulation

strategies, particularly in special populations with altered bile salt levels.

Author Contributions

L.G.A.-A.S., S.P. and K.R.Y. wrote the manuscript; L.G.A.-A.S., S.P. and K.R.Y. designed the research; L.G.A.-A.S. performed the research; L.G.A.-A.S., S.P. and K.R.Y. analyzed the data; L.G.A.-A.S., S.P. and K.R.Y. contributed new reagents/analytical tools.

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Conflicts of Interest

L.G.A.-A.S., S.P., and K.R.Y. are paid employees at Certara UK Limited and may hold company shares. No financial or personal relationships influenced the research presented in this manuscript. All funding sources have been acknowledged appropriately, and the study was conducted independently of any funding body's influence on the study design, data collection, analysis, or interpretation of results.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** psp470177-sup-0001-supinfo.docx.